

Week 1 Assignment Summary: Getting Started with Qiskit

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August 20, 2026

This document provides a concise overview of the work completed during the first week of the Quantum Computing course, focusing on the transition from classical computing to fundamental quantum mechanics using IBM's Qiskit framework.

1. Core Quantum Concepts

The foundational differences between classical and quantum information were established:

- **Qubits & Superposition:** Unlike classical bits (0 or 1), a qubit exists as a linear combination (superposition) of basis states $|0\rangle$ and $|1\rangle$, represented mathematically as $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$.
- **Quantum Phase:** Explored the distinction between global phase (physically unobservable) and relative phase, which is critical for quantum interference.
- **Entanglement:** Mastered the concept of correlated multi-qubit systems where states cannot be factored individually. Prime examples include the maximally entangled **Bell states**.
- **Notation & Representation:** Adopted Qiskit's little-endian notation (least significant bit on the right) and represented quantum states as column vectors and operations as unitary matrices.

2. Quantum Gates and Circuits

Quantum operations must be reversible. The assignment covered the application and mathematical representation of essential gates:

- **Single-Qubit Gates:**
 - **Pauli-X (NOT):** Flips the state ($|0\rangle \leftrightarrow |1\rangle$).
 - **Hadamard (H):** Puts a computational basis state into an equal superposition.
 - **Z & T Gates:** Apply phase flips and phase shifts ($\pi/4$), respectively, without altering measurement probabilities in the Z-basis.
- **Multi-Qubit Gates:**
 - **CNOT:** Flips a target qubit conditioned on a control qubit; the primary gate for generating entanglement.
 - **SWAP:** Exchanges the states of two qubits (decomposable into 3 CNOTs).
 - **Toffoli (CCNOT):** A controlled-controlled-NOT gate acting as a reversible classical AND gate.

3. Measurements and Execution

- **Observation:** Understood that measuring a superposition state forces it to collapse into a classical computational basis state.
- **Born Rule:** Calculated measurement probabilities based on the absolute square of a state's complex amplitudes (e.g., $|c_0|^2$ or $|c_1|^2$).
- **Qiskit Patterns Workflow:** Learned the standard 4-step modern quantum workflow: **Map** (create circuit), **Optimize** (transpile for hardware topology), **Execute** (run via Sampler/Estimator primitives), and **Post-process** (analyze results).

4. Completed Practical Exercises

The theoretical knowledge was applied through hands-on Jupyter notebook exercises:

- **Concept Validation:** Completed 9 True/False and 4 Short Answer questions to solidify understanding of states, gates, and probabilities.
- **Challenge 1 (Quantum Half-Adder):** Examined and simulated the behavior of a quantum half-adder when fed superposition inputs, observing quantum parallelism.
- **Challenge 2 (Toffoli Decomposition):** Successfully derived a decomposition of the complex Toffoli gate using simpler single-qubit and two-qubit (CNOT) gates.